Theory Of Computation
-CSA1303

LAB EXPERIMENTS:

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EXPERIMENT NO 9 :

```
#include <stdio.h>
int n, e[10][10], visited[10];
void closure(int state)
{
   visited[state] = 1;
   for (int i = 0; i < n; i++)
   {
      if (e[state][i] == 1 && !visited[i])
      {
         closure(i);
      }
   }
}
int main()
{
   int m, a, b;
   printf("Enter number of states: ");
   scanf("%d", &n);
   printf("Enter number of epsilon transitions: ");
   scanf("%d", &m);
   for (int i = 0; i < m; i++)
   {
      printf("Enter source and destination: ");
      scanf("%d %d", &a, &b);
      e[a][b] = 1;
```

```
    }
    for (int i = 0; i < n; i++)
    {
        for (int j = 0; j < n; j++)
            visited[j] = 0;
        closure(i);
        printf("Epsilon-closure(q%d) = { ", i);
        for (int j = 0; j < n; j++)
        {
            if (visited[j])
                printf("q%d ", j);
        }
        printf("}\n");
    }
    return 0;
}
```

